

ActInf Livestream #034.1 ~ "The free energy principle: it's not about what it takes..."

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[Music] hello and welcome everyone to actin flab live stream number 34.1 it's december 7th 2021. welcome to actinflab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at some of the links here on this page this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams all acting flap activities are participatory so let us know if you want to join in or co-organize we have three organizational units education communication and tools and weekly meetings for each of those units a ton of interesting projects to get involved in coda.io at activeinferencelab activelab you'll find a searchable and sortable site where you can learn more about the past and upcoming live streams that we have today in acting stream number 34.1 we're going to be learning and discussing about this fun paper the free energy principle it's not about what it takes it's about what took you there by actual constant in and in 34.0 dean and i had a great time we talked about the paper so check out the dot zero and of course read the paper if you want to learn more today in 34.1 we can go over some of the key claims we can look at some of the examples again and differentiate the bayesian approach versus the free energy approach and just really take it anywhere that people think is interesting so if you're watching live just ask questions in the live chat and we'll be keeping our eye on that and we will just introduce ourselves go around and then we'll take it wherever we take it so we can introduce ourselves we can say hi and say maybe something that was exciting about the paper that we remembered or something that we'd like to reduce our uncertainty on by the end of this discussion so i'm daniel i'm a researcher in california and i'm actually just looking forward to seeing if anyone in the live chat has some questions to ask because there's so many cool adjacencies to this paper and just the beginnings of the thread were explored by dean and i in the dot zero so exploring the implications and what people see as relevant in this paper i think will be really exciting and i will pass it to dean thanks daniel

i'm dean i'm calgary um i'm just curious again to see how as you start pulling on a thread discovering how many things are actually there as opposed to just the threads so yeah i think today should be a pretty interesting day and i'll pass it over to david i'm dave i'm live north of manila i did natural language processing with big computers and studied cybernetic learning theory and i'm excited that he's given us some simple math actual axle constant is kind of bringing it down to those of us who stalled by the end of calculus one just to sort of lead-in question dave in the area of cybernetics is there any analogous work anyone who said well it's not really about being a cybernetic system it's about how you came to be that cybernetic system like are there any parallels in the cybernetics literature for the kinds of things that are explored in this paper well there are yeah um in terms of um not so much necessarily the origins well yes in cybernetic learning systems this theory um a branch called conversation theory uh they completely cut loose the notion of the physical embodiments of systems from the mimetic or functional or conceptual or cognitive aspects of systems in fact even types of systems uh very much like what richard dawkins did 30 plus years ago on separating genetics from my medics which in his case originally was just a proof of principle not something he wanted to pursue especially and it has been pursued um in cybernetic learning theory um the individual is equated to the conversation is equated with a s a type of system that originates that persists that merges into other systems that may very well function through many bodies so you have distributed intelligence right out of the gate um as well as internal differentiation internal self-definition of each individual as a conversation definitely the there and there's a lot of math even going way back to the 60s and 70s they they uh there were cyberneticians who took precise quantitative work very seriously and dug down uh very carefully on quantified attention quantified uncertainty resolution of uncertainty and the qualitative consequences of deciding what's really happening what my plan really is very much uh the sort of fun abrupt and often irreversible changes that the world looked at in a lot of depth when catastrophe theory burst upon the world sudden catastrophic perhaps an improvement but a catastrophic a suddenly relatively irreversible change and here it is still working on it in deeper and deeper in detail and wider scope so good for mr dr cost pretty interesting it makes me think about like a friendship and how maybe it's not just about what the friendship is or what it takes to have a friendship but it's about the specific conversations that take you there so let's just kind of look at the main aims and claims and parts of the paper and then either of you raise your hand of course maybe some others will join and anyone in the live chat please just ask any question or make a comment we're just chilling so the center topic that's raised and it's a term that's

introduced in this paper uh not the word but its usage in this situation is the entailment problem and just like the dog and the tail as they say the entailment problem is about the relationship the necessity and the sufficiency relationship which ones following which and under what situations they're following each other the entitlement problem is the confusion in the entailment relation between free energy minimization and life so is free energy minimization necessary for life is it sufficient those are the core pieces of the entailment problem and what happens in this paper is a simple dissolution through a trivial toy simplified example of that entailment problem and we're gonna probably walk through it again today but this numerical example which is very stripped down it's not full active there's no action in the loop it's just a perceptive example and it just shows that free energy minimization does not always lead to the correct choices it's also conditional upon the priors in other words how you got there it's about what took you there because the priors don't come from nowhere they come from the past so this kind of connects this uh rigorous logic within the action perception loop or in this case just a simplified perception loop to the often ill or a logical way that things happened to get the way they are so it's going to be that tension between is this a rational analysis is it locally rational maybe globally irrational all those kinds of topics come up and just to skip to the final words of the paper the claim in this paper that's explored is that free energy minimization is not sufficient for life so not that fep has nothing to say about life not that actinf is irrelevant but that free energy minimization is not sufficient it turns out that it's necessary to be doing something like free energy minimization to persist in a dissipative world but it's not sufficient it can't be all that you have the wheels are necessary for the car to go round and round round and round but they're not sufficient and then that last sentence the free energy principle is meant to account for all kinds of systems you the listener slash reader the author of the paper and the system of interest the organism under study in a unifying fashion so we should also explore like what is the unifying lens or what is the framing under which the research scholar and the person who's just sort of like what is the free energy principle and the bacterium and what is that unifying principle and what does the free energy principle have to do with that okay dean where do you think we should jump in or what would be um a fun question or place to start here well so i have i have an interest a little bit history with this so what what took you there so if i'm meeting somebody and they're an expert at something i don't typically ask why are you an expert because that's kind of offensive but to ask what what got you here what got you to the place where you now bear this title the weight of that title isn't is an interesting way of going about trying to figure out what what logics they have

applied in the past to untangle problems and resolve things that may be somebody with less expertise struggles with her or doesn't seem to be able to overcome so that's where this paper was really helpful to me instead of why do i have to do this or why are you such an expert asking where did you come from and then when being specific in terms of how did you resolve something that i still find difficult to resolve like why do i have to tie my shoes and then the expert says well because you're going to trip over your shoelace and fall down well i haven't tripped over my shoelace instead of going down that path figuring it out in a slightly more sophisticated way cool thank you um i'm not gonna read this blate quote but of course we have to write it down it makes me think about how why which is an important question often lends itself towards generalization like why are you studying machine learning well it's really important and it's going to become increasingly important and i wanted to have a good job and other people told me it was relevant it kind of generalizes out whereas what took you there is always going to be a trajectory it's kind of a path and that's where you get the answer that's more like well i met this mentor and then i was curious about this specific topic and then we did this program what took you there is about the specifics and is broached in why questions is often generalizations and we know that there's not even a single answer to why like we've talked about aristotle's whys and about tin bergen why like why is always going to be a plurality of answers whereas what took you there is going to be a specific path and so that's the tension in living systems is we want to understand something general like surely there are some generalizations we can make across single-celled life or across mammals but also we're explaining specific systems which aspects are going to be generalizable and which ones aren't and i think again free energy principle as a theory of everything and something that does clearly make generalizations across systems at least apply to different systems how does this hyper specific trajectory based way of approaching answering a question relate to more of a mean field approximation averaged over multiple categories i think it's that tension that's really interesting to describe so one is this general versus specific tension and then if we can go to the simple point that's made in the paper that the free energy principle is not concerned with the sufficient conditions of existence but rather with what must have been the case given that you exist so this is related to necessity and sufficiency which we talked about in the dot zero they're similar words so we should use them differently they're different words in other words it's about necessity but not necessarily sufficiency and then it's not about what figuring out what it takes to be alive it's about figuring out what took you there the title is restated and turned around many many different times because it is such a distillable and simple yet

important concept and i think the piece that i'd like to explore a little more is really the pronoun use i understand why the paper is written from an i perspective because that's convention with single author philosophy papers it's about figuring out what took you there so there's a sort of rhetorical element of speaking to a reader which is returning at the end of the paper those last few lines about you me and the system under study but then also it's almost like about how the free energy principle is about systems looking at their own timeline and trajectory and perhaps not as much as looking out so i'm curious about what does that relational element mean here with the person relating to their past and us as systems relating to other systems dean yeah so the necessity might be you show up you attend in order for things to sort of continue on but the sufficiency part is how available you are so you can show up and turn all your senses off you can be present but not necessarily be taking in anything or at least trying to actively avoid taking information in and so that's the part again i would agree with you that's the part i found really interesting because to not understand or appreciate that there's a difference between what is necessary i'm i'm here i'm in part of this live stream now but then i could be distracted in all different directions not really be able to be available to what is going on around me is is a whole different matter and i think that's what axel has done a really good job of bringing it down to that personal level sufficiency has to be exhaustive to say what's truly sufficient for the bacteria to live it's going to include things like well you need the laws of physics and you need gravity and you need um just everything it's like really you're giving almost a bit by bit play-by-play so sufficiency it requires an exhaustive account because if you're missing something that's necessary you didn't capture what was sufficient necessity allows us to just focus on some aspect that we know is relevant and then talk about the why and the what and the how of necessity while acknowledging that we're just focusing on one piece like that one line in that one kandinsky painting it's necessary for my interpretation of it it's not sufficient because the sufficiency of the interpretation of art would just send you off into the whole universe maybe you need all of human cultural history including from cultures that you wouldn't even directly acknowledge for the sufficiency to be garnered in that interpretive setting but necessity is a lot easier to point your finger to that relates to what are distinguished as the strong and the weak responses to the entailment problem so let's look over at this and again anyone with any kind of a question please just ask it because we're chilling and looking forward to your questions so this is where in the paper the entailment problem is specified arguments in the literature on the free energy principle give the impression that in order to be alive to count as a living system one must minimize free

energy and so there's a few ways to think about this relationship between minimizing free energy and life sorry if you thought it was going to be a discussion on a different topic because that's what this paper is about and it just again it's so simple it's like a shape in your hand you're turning it over the shape is simple in some ways but then the implications of the shape and the way it gets connected to other ideas is really critical actual distinguishes two ways to talk about the relationship the strong and the weak and as we'll see they differ not just in their strength but also essentially in the direction which the arrow is pointing so it's kind of like it's not like there's a little mountain in a big mountain but the arrow of elevation is pointing the same way there's something a little bit different happening the strong claim in response to the entailment problem is that minimizing free energy is a sufficient condition for life and that's also called in some earlier work the overly generous claim which is kind of a framing that already says that it's probably too generous it's incorrect in that way and the strong claim is the sufficiency minimizing free energy that's all you need it's sufficient to minimize free energy to be a living system the implication of the strong claim would be that all we need to do is have that free energy thermometer and then we can determine who's minimizing free energy we have our meter stick on this left side and then that would be sufficient to talk about what systems were alive or not in contrast the weak claim is a necessity claim if a system is currently alive it means that it has minimized its free energy and such a type of claim does not assume the fep is designed to set the bar for the sufficient conditions of life or meant to predict what things may or may not be alive rather it limits the scope of the application of the principle to beings that we think are alive now and enables us to know the necessary conditions under which those beings can be living i.e. can actively resist the loss of structural integrity what took them there's a few interesting pieces just they differ in strength these two responses or claims that are generated by the entailment problem i wonder what is the scope of the fep and is it for systems that we have a priori decided are already alive are non-living systems under the scope of the fep so maybe we can try to like do some duplicate the slide and do a little diagramming and venn diagrams on does the minimizing free energy circle look like with respect to the life circle and then what is life if we don't have the thermometer over here on the left side so can i ask a question daniel would it be going too far off the rails if i if i wondered to take this idea and say okay so i want to build up i'm i'm an academic or i'm a trainer and i want to i want to build out a curriculum knowing this what does that imply in terms of well i know what's the difference between something that's surprising removal because i remember i'm the one that already knows this and i'm trying to pass this along to

somebody who doesn't know it so it's i'm i'm trying to guess what surprisal removal or minimization means to them i'm supposed to be guessing what life means to them i'm supposed to be knowing what is necessary what absolutely they must know and what will be sufficient so in that context how would we um how would we order this let's kind of lay out a few options okay so we have um behind the first door we'll use letters to talk about them so we'll have i'll take the cash monty are you sure i can reveal that one of the doors has nothing behind it okay so we're gonna have scenario a which is their um r systems that are minimizing free energy so they're systems outside of that circle so there may be systems that are not minimizing free energy but within the set of living things all of them are minimizing free energy so let's think about what kind of world that is so then we can have scenario b where there's living systems there's non-living systems as well so we could ask what puts us in one category living or non-living but there's living systems that are minimizing free energy and there's ones that are not i think we can say that because if we go with the weak claim then it's fair to say that b is ruled out because this would say that there's living systems that are not minimizing free energy so we can say like basically weak claim uh disreputes this area the area of that crescent moon that's being um sort of invicted against uh is distributes a word disputes we'll go with that the weak claim disputes this area of the crescent it's saying well all living systems have to be minimizing free energy it's a necessity but it's not then a third scenario would be like they're the exact same thing so well that would be sort of a well you could have minimizing free energy is a sufficient condition for life so that means that the life circle is we'll have to we can't show total overlap but the life circle everything that's minimizing free energy is alive so here in a there are free energy minimizing systems that aren't alive like a ball falling off of a cliff might be minimizing its free energy just like it's minimizing its potential energy or a burning candle might be minimizing it doesn't have to be alive so that's these are free energy minimizing systems in the crescent of a that are not alive in c we have this perfect relationship between the two things and that is a very strong claim which is that if it's minimizing free energy that's sufficient for being alive and if it's not minimizing free energy it's not alive this is sort of the the weakest claim is on the left side doesn't mean it's useless doesn't mean it's inadequate but this is the weakest claim this is the necessity claim then we have the strong claim here that minimizing free energy is sufficient well this is this is the strongest claim in c that minimizing free energy is sufficient for life so anything that you know if doing x is sufficient for greetings blue hello cool light we're um in the slides on 24 just talking about necessity and sufficiency c is the strongest case it's saying

it's sufficient and so you're either going to be minimizing free energy and therefore sufficient to be called alive or you're doing neither and then b is a little bit of a middle case that's saying there's some living systems that are not minimizing they're free energy but some are so these are three now are there other possible layouts and then dean let's return to your question about like educating and how do we think about this how do we distinguish these scenarios in an education context or welcome blue if you would like to say hello or just what makes you excited for this paper um i loved this paper i thought it was great um and i have been listening along in the live stream so i'm kind of up to speed with where you guys are and what you were talking about but um i yeah enjoyed axel's writing and um i like like the fact that i can implement the free energy without having to ascribe to pan psychism and also without having to ascribe to the mind life continuity hypothesis or all of these things that we've been discussing um like the necessity sufficiency it's there like i think that axel sets out a really logical argument or a set of arguments doesn't make any claims that aren't um justifiable with maths yes using a simple mathematical example um we'll we'll keep this slide around because we can wonder maybe is there some well another relationship that we could speculate would be like a kind of something like this there's systems that are minimizing free energy or not there's systems that are alive or not and sometimes they overlap but they don't have to overlap but there's some special living systems that are doing it and there's some that aren't and there's some that are neither so this is the most uh we'll just call this one situation d d is the most is the most flexible scenario because you could have a or b i mean well i won't use those letters you could have life free energy minimizing both or neither so d is the most flexible case c is the most restrictive case and then we have two possible nesting relationships where one is nested inside of the other it's kind of interesting to wonder what kind of world do each of these exist in um well where is the mind bubble yes let's should be around the whole slide is it is that the mind the the modeler is that why there's that relational or personal framing of the paper because the mind is the mapping of these different venn diagrams dean and i think it's really interesting too daniel that in real time we now have a slide 23 and now a new slide 24 and although the we now know the order that they they presented themselves to us as could we have started with 24 and worked our way back to 23. that's that's when it gets really kind of weird and twisted but it's also what i think the paper is speaking to if we're actually thinking about this and as a as another example of a post potentially post diction way of viewing things historically now yes so 23 which now has a little logo on it for those who don't have the slide deck at hand 23 was derived heavily from the paper right it was quotes from the paper and

then just a visual representation now that was not in the paper but it could have been interesting and then that's right through the specifics of who joined when and what we were thinking about when that's what led to 24 but maybe we'll add another slide before 23 and then they'll both be a different number it does bring up an interesting question i think we could look at well first dave thanks for writing in the chat do you want to read what you wrote or would you like me to read what you have in the youtube live chat yeah um could i expand on it just slightly please what i would have posted if i didn't have the big limit title necessary versus sufficient two of the original concepts of general systems theory have recently as of 1975 been resuscitated there's an admirable discussion by russell akov the first of the two concepts is that whereas causal argument considers necessary and sufficient or double implication conditions systematic theory or cyber systems theory or cybernetic argument deals primarily with a logic of necessary conditions see fred singer's discussion of the producer and product relations and in summerhof this is called goal relations and i'd add this deals with intentional action where there are goals there's an old painful discussion going back to way before darwin that the world exists only as push galileo wanted to absolutely banish goals of any kind any value had to be absolutely driven from science if you say for the purpose of is get out you're not a scientist uh i continue this notion underlies the more sophisticated treatment of goals and intentions conceivably partly specified intentions fuzzy goals goals with um ill-defined uh criteria which you might or might not be aware of i would the other concept of about the same importance is the presupposition of a systematic universe there is a tacit assumption that things objects and other elementary entities are interdependent rather than being isolated units which is the assumption behind the majority of sciences further as a result of their interdependence or with the same meaning of the supposition that things objects and so on are not really unitary these entities form systems and it is systems which may be observed and manipulated so the notion that there's one and only one way to approach an argument one and only one order among real things that are scientifically respectable utterly wrong utterly wrong if you can't work your argument in at least two different directions you haven't thought about what your argument is you don't know the boundaries you don't know what it is you don't know what a jellyfish is you think you know what a jellyfish is but you don't know the difference between the inside and the outside you may end up in the inside so it's almost like if there's only one to run through the argument that's when necessity and sufficiency are easily confused so if somebody just drives in one lane from the beginning to the end and i made my case and that's all i have to say well then you've implicitly through your actions your enacted behavior you've

uh acted consistently as if what you had to say was sufficient for making your case the prosecution rests and also it was necessary you think i've included everything necessary and sufficient so that's actually where you get and here's maybe where we should duplicate the slide again you get that's an overlap of necessity and sufficiency but then if you did two parallel arguments just you did one from start to finish and the other one from start to finish now you could say both of them in their own timeline they were both necessary even if they have totally disjoint information that was being used and then what is the relationship of the goals there and that is kind of like dean's multiple eyes open giving us the perspective to look at goals when we have multiple realizations because it does help us distinguish necessity and sufficiency because we literally see well it could have been done a different so it being done in way one can't be the whole story because we just saw it what done in way two i'm sorry dad you dropped off on me for a second there so i wanted to ask david a question about yeah dean kind of drops in and out he was giving some funny descriptors of his internet hardware earlier it was pretty funny start again sorry um so i agree with with what david was saying and there's a there's a book called the origins of order project and system and what the author basically says is that projects are easy to get trapped in goal directed behavior whereas where if you can also hold up the system you can see the number the sort of the multitudes of different directions all at once and so i was wondering is is that does that sort of fit in with what david now was trying to speak to try to find out the author's name here well stewart kaufman uh who is i think he has become a little more active recently i know they're pushing his book origins of order really hard that's something i read in like the 80s who's going this is a new one this is paul khan yeah okay k-a-h-n yeah k-h-n and he's and he's basically trying to use the argument that you can look at the constitution or you can look at the actual laws and and you hold up both one is goal directed the project piece whereas the other one is is is trying to work from within and around so that's it's that sort of between and inside argument and that's not the same as as projecting yourself to a particular outcome yeah i can't i can't answer that right now but i do want to tell a little quick story uh when i was working as a senior programmer analyst i was organizing this project it was going to take a couple of months and my super is kind of looking over my shoulder and i have the criteria and i have a big flow chart and he says oh so dave this is how you're going to do it and i said i haven't decided he said well wait a minute what's missing i mean you've you've got the whole thing laid out it looks like it ought to work and says yeah i haven't decided i've only done it one way there's no decisions involved in that i've just stated the requirements i didn't decide how to do it so then he went out and had a smoke i

ended up using a much better technique that saved us a huge amount of rewrite that nobody would have noticed because they would have said oh yeah the answer go for it it's almost it's the zero to one which we've talked about and it's also related act infer serve infer you'll have something to infer when you've acted whether getting that first draft out or just doing a private voice memo or whatever stigma g you need you'll have something interesting to infer if you act i promise and then maybe you'll be closer to service than you expect but infer is still within the realm of like well what would be sufficient to get the job done but if you go well what would be necessary to do this one way this is what would be necessary we could um walk from california to new york that would be necessary for getting to new york okay it's gonna take too long i just don't have that kind of time what would be another way to do it so there is a almost a necessity and a sufficiency action and inference where action has a lot more to do what's needed and inference or thought maybe moves us more towards thinking about what's sufficient not sure about that but also thanks for sharing this um this resource well and back to what david was saying i in the work that i was doing i often had people who who are very well versed in setting up all kinds of sophisticated projects asking me where's the project in this and me having to jump up and down and wave hands and say this isn't a project that's why we're doing it we don't know how it's going to turn out until we do and then we can look back on it and say well did we minimize free energy we weren't using minimizing free energy but because i wasn't you know first in that lexicon yet but that's essentially what we were doing we wouldn't know until we'd arrived that we'd arrived and that's that's really uncomfortable for people who were constantly saying show me the milestones okay well i can show you the stages and i can show you how identity identity is now in turning into identifying as people are are constantly processing who they are as they're on board this this this idea but i can't tell you what it's like because it's not necessarily goal directed other than the fact that we want to try to keep it open as long as we can and figure out what logic people are using in order to be able to untangle and pull things out of hats that that others don't know how to do so yeah one other thing that makes me think about is grants applications so a grant application might frame things in terms implicitly or explicitly in terms of sufficiency if you fund me we're going to fix this disease like this is the grant that will be sufficient for fixing this disease whereas saying that it's a necessary step like we do need to understand how protein one two three works it'll be necessary but it's not the whole picture it's not sufficient so so much when we really turn it over is related to because there's so many things that are and so little save for totally exhaustive examinations of really contrived systems so rarely do we find sufficiency because

if you can define sufficiency you can make that bacteria in the test tube de novo and then life it's kind of like puts a little twist on it like what would be necessary and sufficient for having a bacteria in a test tube putting a pre-existing bacteria in and having it replicate so life as this paper explores it sort of threads the needle of necessity and sufficiency because the past is necessary and sufficient for living systems to be in the state that they're in now and so that's like the generalization as well as the minute particulars if anyone wants to ask a question otherwise i think it's good to take a second look at the numerical example yes blue first so just like what got us to this point the past is necessary but not sufficient but it just just echoes the title like how did we even get here right so let's look at that bacteria and then of course we can jump around anywhere else okay so the numerical example is of a sensing and feeling a sentient bacteria not necessarily experiencing and action isn't even in the loop here but it's a sensing bacteria the organism is inferring its prior beliefs about the cause of the observations it makes so it has a prior let's just go over the um sort of letters again so there's a class of events for the organism so this is like we need a special color or some metadata tag like how can we tag map and how can we tag territory let's just use like one set of colors or one font like map comic sans territory some calligraphy script or something like for the organism of the organism now it's our map of the organisms map a and b are events that are part of the class r and so r is like the receptor state are inferred downstream of a chemical signal which is s so s is like for signal and s is whether the actual molecule on the outside is the a or the b so this is kind of a discrete binary choice it's briefly discussed in the paper like what happens if there's multiple molecules or one could imagine that what's being inferred is a continuous variable like blood sugar but alpha and beta are s that's the outside stuff and then a and b are the organisms receptor inference blue go ahead and then i i'll just both end yeah so just to sort of look at it in figure one or in the formalism uh block that's in the paper the organism has a prior and it's the specification of priors that differentiate bayesian statistics versus frequentist statistics among a few other pieces but that's really the key piece and so the prior p of r is 80 20 in favor of a a priori before how did you get there but a priori it's 80 likely that it's a and 20 b before any info comes in then there's a likelihood mapping so that's the the distribution of the signal conditioned on the receptor state so we're talking about what's the probability of alpha outside conditioned on a and b and so it's 70 a and 30 for b and vice versa so what's interesting about this is that's a pretty noisy signal it's really only saying it's like if it were 50 50 it'd be a useless signal because it wouldn't knowing it one way well it would also depend a little bit on the frequency of alpha and beta but a very noisy signal like 30 of the time the

receptor is giving the wrong answer so this is a way to show that with multiple uncertainties connected the right way you can actually tighten your estimate because in this example with the 80 prior on α and then a 70 likelihood mapping which is also a that's more like classic active jargon after observing α it turns out that the estimate of α the estimate of state being a conditioned on α being observed tightens that prior from 0.8 to 0.9 there's still a tightening of the prior by a significant amount even though the mapping between the molecule and the receptor state is somewhat noisy so that's like a bayesian bacteria it's just applying bayes rule to this simple binary outcome that it's doing inference on blue any thoughts there yeah so uh you know i this goes back to something that dean brought up in the dot zero like is oh and it's actually something that axel wrote in the paper like is the prior subjective or objective um and like the mis dysfunctional requires and the survivability of organisms with dysfunctional fires so i just feel like i mean i know axel says it's subjective but i feel like it's nested so i feel like there's the prior must be a subjective prior over and objective frequency right like so so if i frequently observe you know sunshine outside when it's warm or something like that like if mostly when it's more when i feel warm i absorb sunshine so when i have these two things linked like based on that frequency i then make a subjective assumption and and that's how i feel that the the structure of the prior um kind of has to be otherwise like it doesn't make sense to put the math into the prior um like like in my mind like if you can't if an objective prior is based on frequencies but a subjective prior is not a subjective priors i guess but it's a guess over the frequencies i don't know i feel like there's like an extra layer that needs to be tucked into the subjective prior if you're gonna put math there one way that that's dealt with empirically and then dean is what's called parametric empirical base and in parametric empirical bays you kind of start with just the prior being what you observed in the past so if you had eight sunny days and two rainy days that's where that point eight and point two prior comes from it's um not a way around specifying a prior it just starts your prior in a place that's empirically grounded yeah um so i think in the point zero two i was i was trying to figure out how the likelihood is really the kind of background that we have and that's what blue is kind of talking to you and i was wondering about in terms of context because if i see a black dot on a on a white background i think of period whereas if i see a white dot and a in a in in black background with a whole bunch of white dots i think how many light years away is that concentration of hydrogen right so again we could d we could decontextualize this but if we do d decontextualize and this is where it gets really hard for me i get lost in all of the terms but if we do decontextualize it does that does that help us in any

way understand both the subjective objective factors in play that's all i'm i'm wondering about and again i get i get lost in the number of different ways to be able to describe this that's part of my problem i have an upper limit on how much of this stuff i can do once if you want me to focus on days i can do that that's hard enough for me to concentrate on that but if you actually think about this okay so what does this actually imply in terms of entailment then it gets really really tricky because i can't just focus on one thing at once i gotta sort of think about what all the implications of that are and man can never go off in different directions all at the same time so i don't want to be too tangential here but yeah that i i'm still not sure whether the subjectivity in isolation can work i'm not saying it can't i don't know thanks dean blue then dave so in like the empirical parametric base that would be like an objective prior like do you have subjectivity in empirical parametric base because like that's what axel was describing in the paper you do you can have an objective prior based on frequencies like it's 80 of the time you know sunny and 20 of the time rainy and so you you get these frequencies and that's what goes into constructing an objective prior but but when it comes to like an organism and a choice or even like a scientist conducting an experiment you know like why would you even do the experiment in the first place you have a guess right like you have something that you think will be the outcome and so like a totally subjective prior is like well i think sixty percent of the time it's going to be blah blah blah and like that's a completely subjective prior it's just your best guess which which if you're gonna try to really do objective bayesian analysis like you need to have a flat prior not your best guess right you need to have your prior needs to come from previous observations and not you guessing but but you could very well like not you know do a very empirical or objective bayesian analysis and just say my guess is this and then you count make the observations observe the frequencies you know you go through the entire time step and then at your next time step there is that tightening between the likelihood the previous likelihood and the prior so that that is um i mean as far as i understand it now i'm definitely not an expert but so i think like the subjectivity and objectivity i don't know i think that there's some some link there like i get it for scientific bayesian data analysis like you want to have a flat prior you want to do objective analysis you want to use you know the previous frequencies or whatever but i think that um you know in terms of a mental calculation it's also over frequencies or it's your best guess like what do you think will be the outcome and it's not always doesn't always line up right thanks blue dave the term decontextualized i i kind of that shocked me when professor jaron started using that it's just he contextualized what that that's just some some jerk opens his mouth and blood he contextualized his feelings at me and you

know if i want to know what's going on i hyper contextualize i've been particularly attentive to my own prejudices and my own viewpoint you'll think about what goes you know einstein hated the word relativity he wanted to come up with something like hyper contextualized or the real world or the the um the tensor description uh where you don't have to ask where are you because you have the structure of the world what do people mean by by d contextual odds uh is it is it a term of praise as well as a term of contempt or have i just always misunderstood so i love this term hyper contextualized i i don't know that i would use it in the way that you use it dave i i would use it like i would play out all of the potential like what ifs like like let's talk about counterfactuals can we can we put it into you know how many different contexts will it fit um and that's when when i think about hyper contextualization it's like playing out all the potential scenarios for me thanks dean yeah and building building on what blue just said so i i saw a youtube video where a guy figured out a way to make a target move so that no matter where you threw the dart as long as you were directing it towards the dart board the target would create the algorithm would create the target moving so that you hit a bullseye every time and i was a phys ed teacher i was like oh this is amazing and i took it to one of my colleagues who shall remain nameless and i said we should show this video to the kids and he was like why and i said well think about it every time a person moves their hands to catch the ball they're making the same kind of adjustment and it was like a light went off a lot of the things that i was that i was coaching were then decontextualized and the proof was in the fact that when you saw something out of context that was running the same parallel function you didn't necessarily recognize it so this was a really i i love hyper contextualized so i wonder if that's like abstraction or d irrelevant contextualizing you take away the irrelevant part of that setting which is oh it's hands or some other specific system and then that paradoxically gets you to hyper contextualize because highlight the parts of the context that matter dave and there's that famous phrase from 1923 consciousness of abstracting training people to think scientifically one of the first things you do is get them really aware that they're how much they're missing you're abstracting away from at least 90 of whatever you're looking at uh and since i've got the mic um there was the question of how a fielder catches a baseball a fly ball here it comes it's coming out of the sky at you and you know the good fielder knows how to do that you ask him how you well i just i just go to where the ball is going to be but he can't tell you how and there were a bunch of people mathematicians and programmers and physiologists and philosophers trying to how in the world are they doing this it looks like they can't because we can't come up with an algorithm that doesn't and they finally watch them and

i guess they got one one one guy that's high on whatever it is openness the big five uh and and they says well what are you doing how do you know that you're gonna catch the ball oh i i know i'm gonna catch the ball when i look at it and it doesn't move it doesn't move yeah it just hangs there and i put up my my mitt and it since it's i know where it's going it's going to my mitt and it's not moving and it's in my mid and it didn't move then either that's how they get there they figure out what point in the field makes it look like the ball's not going to move as it comes toward there and that's where you go and then just hangs in the field till it's in your mid classic baseball physics the thing i thought was have you ever been high on openness but then it's true there is a simple algorithm like if you're looking and it's to the left then you should go to the left if you're looking it's to the right and then you're cool active baseball and anticipation and the path of least action you want to talk about high on openness i don't know that it reminds me of our conversations with adam saffron right like what the psychedelic like i would like to see like the psychedelic like comparison with big five like the integration of those two kind of things in terms of active inference because i do think that um the openness like increases a lot under under the psychedelic influence and so it's um not have you been high on openness but has being high ever made you more open maybe is a different question and well a few things there's sort of the um rebus sevis albus relaxation of beliefs under psychedelic strengthening or just the altering of beliefs with fristen carhar harris saffron and what is being what priors are being opened which priors are being tightened and that was sort of at the heart of the rebus versus cebus and then the sort of um hegelian synthesis in albus and then also in terms of how we got here the first two weeks of january was live stream number 13 with adam and colleague on the cybernetic big five and so that actually sort of started our year with thinking about how frameworks that a priori have nothing to do with actinf or fep like big five was a principal component analysis based variance decom uh various partitioning on psychometric data yet we saw how active could help us reimagine that and generalize it while also bringing action into the loop so then it's kind of cool to see how the personality traits come back into play uh dave and then we're going to return to base and then we're going to move into the variational free energy well stop me if i already mentioned this but um there's been some criminological studies of the way psychedelics are used by as a recruiting tool of violent cults you go to a rally in favor i will not say whom and if somebody seems like they're really grooving and really into it you give them a little ecstasy and that locks them into the mentality and if he's kind of getting fired up but but he's not quite with it you take him home give him some lsd and take him to the session and that moves him out of what he was in and

moves him into where he needs to be in order to do the right things in life maybe another a little bit of ecstasy now and then over subsequent months to just get him locked back in where he needs to be just like just like safran was saying you either get him out of the of where he is or you get him right back right deep into where he is well there's a long history of people engaging in experiences taking substances to move their psychodynamics one way or the other um i think the active streams verge on it but we also remind people to be responsible and legal with their behavior and with the one and only brain and mind that they have let's return to the bayesian maths and then see how that bayesian bacteria moves into the variational free energy bacteria but i pulled out that one quote from a um online statistics resource so we can you know uh the 0.0 is the frequentest bacteria and so the frequentist is basically looking at the outcome of an experiment you flip the coin ten times it comes up seven heads and three tails now that's not the most likely possible outcome for a fair coin but it's totally not an unexpected outcome in that case the frequencies can produce a 95 confidence interval given the variance granted by the sample size and say okay if i get um within these two numbers um then i'm going to fail to reject the hypothesis that it's fair if i'm outside of those two numbers with my empirical results i will reject at a given p-value a given alpha and so for the frequentist there's no sense in asking about the probability that the coin is fair it is either fair or not fair so that's like accept the hypothesis implicitly because of the failure to reject or reject the null hypothesis in favor of a specified alternative the frequentists make statements about the probability of a sample after making an assumption about the population parameter which is just the kind of average out approach average out value of the probability of tossing a heads the bayesian in contrast starts with information about coins so specifies some distribution so it could be like a bell curve a gaussian centered at 0.5 it could be a super tight prior could be a very loose prior and then the extreme case of the gaussian with infinite variance is the uniform distribution and so that is like a flat across all the possible values that probability assessment of the heads proportion is called the prior probability each person might have their individual assessment based on their personal experience called a subjective prior alternatively a prior probability distribution might be selected on good properties of the resulting estimates called an objective prior i don't know if this is the only way to use subjective and objective in the bayesian context but this is just an important difference between frequentism and bayesian so here is our bayesian bacteria again it has some a matrix that's mapping the signal conditioned on the receptor state and then it has a prior over those receptor states that bayesian bacteria then again it's a bayesian perceiving bacteria we're not bringing action into the loop yet but

we're just assuming like if you had the right estimate then it would be easier to know what to do correctly um this is called exact base because the exact numbers are plugged in and then there's some exact number that comes out and it's pretty much just granted by bayes rule what's the issue well it's unclear whether living systems have sufficient computational power to accomplish that especially when there's just massive multivariate landscapes what is the answer in the fep instead of doing exact base there's what's called variational bayesian inference so here we're going to propose this q function and the q function is going to be from a family that is specified a priori and it is more easily fit than just sort of the exact base and then the way that we compare is not by comparing at the end the posterior likelihood but rather we can compare different options by comparing their and this is how free energy is being defined here it's the negative sum over r those are the receptor states of some q distribution and then the natural log of the joint distribution of the states r and the signal α multiplied uh then divided by q_r and so then in dot zero we explored a little bit about how that's like the recognition model that's like we're going contingent on α and now we're doing a hyper parameter estimate like α we're going to a and then what we're not going to be exploring here is like a generative model and then we talked about how those map onto perception and action in the world so then the numerical example shows that okay so the bayesian bacteria it did the right thing it didn't go to that rally that dave was telling it to go to but we can also have a free energy creature bacterium and we're going to see a case in which the free energy minimization does the right thing and we're going to see a case where the free energy minimization does the wrong thing the difference between these two cases is going to be the prior that's set and so that is the crux of the argument that minimizing free energy is not sufficient to be alive aka make the right decisions because both of these examples are going to minimize free energy and one of them is going to minimize free energy in the right direction and make the right choice because it had appropriate priors a maladaptive priors can minimize free energy but still result in the wrong outcome so in this case we have the free energy calculation for um this 80 20 prior with a favor of a contingent on observing α the free energy of this is 0.47 which doesn't have an interpretation as a probability importantly so that we're not within that zero to one bounded space um in contrast the representation of b so that q_r setting that to b that has a free energy of 2.2 so free energy minimization saves the day in that context however it's not simply because the free energy was minimized that the right decision was reached here is a creature with a prior on b of 0.8 we see that just barely upon observing α the free energy of representing b after sensing a is 0.96 and the f when representing a

after sensing α is 1.094 the ensuing posterior likelihood is 63 percent in favor of b and 37 in favor of a we sort of did move the needle towards a because we observed α but our prior was so far off that just one cycle of free energy minimization still results in the wrong that's the crux of the numerical example provided in this paper and the only point that it's being made not this is how you'd model a bacteria with the fep the point it's making is free energy minimization is not sufficient for life so daniel i got a question to ask so i don't know because we're just talking about sensing we're not talking about you mentioned doing but i is it is it is it doing or is it directing so because of the sample sense and because of what the math points to should we think of this kind of analogous to a compass we're pointed in the right direction we still haven't actually acted or developed a feeling from the sense yet we're just we're still pre doing we're we're we're orienting is or am i wrong so if anyone else has a thought after this so let's let's i think there's two few aspects to this compass and orienting approach this little toy motif about the recognition model about having a prior overstates in the world and then getting some input that could be plugged into an action model in a slightly more elaborate case so for example as written in the caption for figure one more biologically realistic descriptions of behavior require discussion of active inference they require it it's necessary and sufficient you just have to do it behavior is the result of a different inference process that of an action policy so π is not here in this bacteria this involves more priors namely about the transition between hidden states and often about preferred sensory outcomes so inactive inference with a partially observable markov decision process we've seen how there's like a matrix of the transition frequency b between the states at one time and the states at the next time and then policies are inferred based upon which policies minimize free which ones are going to have the best chance through time of bringing states into alignment with preference the other thing that it made me think about with the compass is like the exact base returns u_m a probability and that is sort of like one compass needle giving you just saying okay you're going north by northwest or whatever it happens to be this is your one estimate you could be wildly off but given where you were and then just one blip of information that's where your compass needle is now pointing to free energy it's almost like there's two compass needles in this binary and then we're going to select which one has a lower free energy because we're looking the fact that this is 0.478 f and this was 2.27 they don't have a natural interpretation just like entropies you can talk about the difference in entropy but something having like four doesn't have an interpretation in the and so it's kind of um maybe like there's two different compass needles and we're looking at the difference between them or we're going to select the one that has a

lower free energy yeah i don't want to say that orienting isn't a form of doing as well it's just that what the what this shows what this shows me is that like you i you i cut out again but when we're when we're trying to when we're trying to determine with a background um we haven't taken an action step we haven't developed a feeling yet do we do we categorize that's differently than if we were to look at this through an active inference lens we would then push a policy forward right so i'm not sure it is the orienting fitting you fit into the policy piece or is it pre-policy that's what i'm trying to get clear in my mind good question it makes me think about the uda observe orient decide act loop and about our recent paper that connected active to the uda stages orienting i think in this reduced example the um the north pole north star whatever it is is the prior like that is what is oriented relative to so that's a purely recognition model sense of orienting whereas in the physical realized world orienting has much to do with action and wayfinding but this is sort of in the recognition there is still an orientation of the prior but i'm not 100 sure what do you think what i what i find interesting is is that when i look at this and i see a like a 0.8 and a 0.2 it's very clear that we're somewhere between 0 and 1. but when we're actually getting to that wave finding space and we're talking about entailment one whole over two wholes is different than two over one so as soon as we move past one into some of the more as you say material realities that we have to face down that's when those actions become material like we can see what the process ontology is but what axel did was he brought the process ontology down to it even even closer to zero point because he was using free energy principle and he's talking about free energy principle and bayesian stuff so again sometimes i have a general sense of this stuff and i just enough to get me into trouble and be an insurgent and point it people who are really really smart and kind of go i think i know what you're on to but it doesn't get me deep enough to to be able to write a parallel paper to axles and sort of either build them up or or tear them down so yeah i don't know i'm just that's why i'm asking i don't know where oriente or fits in this because i think it fits in the zero to one space and i think it fits into the one to three space as we as the dimensionality blows out that's back to the subjective crossing back and forth and again i come off sounded like a complete idiot here because there's somebody way smarter than me who can blow my argument up but at least i'm putting it out there i'm showing up oh you're you're by being on a live stream and talking about it you're like 80 of the way to the paper dave is there something in this context that would correspond to urgency and i'm particularly thinking in the terms that freud was concentrating on when he talked about how to get people to be less neurotic he says if you're the the point that a person is in real trouble as a neurotic is when the dominant factor in his life

is reducing his anxiety and what we want to do is to get him to tolerate his anxiety tolerate for instance ambiguity not knowing is my wife mad at me or not but don't get crazy sometimes you just can't do anything about it calm down tolerate the anxiety is there something and i i'm just taking that as a case of urgency urgency seems like something that would fit into this literature better so i just want to um like so i don't know about urgency i'm going to kind of derail off of that and more hook onto this toleration of anxiety or toleration of circumstances that are not preferable for like a later more preferable outcome and so this is something that is taught like i don't know in self-development like a lot right like you have to put yourself out of your comfort zone but i mean also like in like the meditative traditions like to learning to like be with that discomforting feeling and not like you don't have to like get all upset and do anything about it um so i i just wonder if that fits like it's a different different thing different than urgency but what about like where's the temporal there has to be like a loop right like uh i'm gonna tolerate this bad thing for something that i think will be like a more predictive outcome later like how can i bathe that that would be like an interesting mathematical um thing to try to work out cool i added a uh slide here here's the eisenhower matrix the x-axis is urgent to not urgent and then the y-axis is important to not important so one of the reasons why this is such a useful and well-known tool is it separates urgency from importance because there's things that are urgent like a chat message that urge you to do something or they have sort of a short time horizon but then there's things that are important or not important there's an orientation when you're figuring out what is in what quadrant kind of like the rumsfeld matrix as well has implications for action so let's see urgency and anxiety dave you mentioned i don't know about the freudian urgency concept much but it's of course with uh psalms with fristen and the project for scientific psychiatry there's some clear links there so it wouldn't be unexpected the idea that the imperative is to reduce our uncertainty through updating our model learning and through action so the reduction of anxiety is in imperative and then sometimes it crosses a threshold towards being truly urgent so it's kind of always in the background but then other times it really is urgent but it's always important and then blue i agree delayed gratification it's like having the time horizon depth to take the right policy selection today so that your expected free energy is higher in the future so it's like would you rather have one dollar every day a thousand dollars in one year depending on how badly you need the money today and what you think you'll do with it in the meantime and all these other things one person given their priors might minimize free energy and come to two different conclusions and i think that's where we return to this subjective versus objective fep take

which is like a person could rationally evaluate given their subjective priors and choose to take that marshmallow maybe that is just the waiting that they have on their priors or maybe somebody does the opposite and that is just where their priors and that stimuli that's the action selection that they take not sure how all these pieces tie together i think it speaks to the the the logic of action and inference and we want to have a logic of action and inference or a framework for it broad enough to accommodate like making bad decisions if our action and inference framework is only tightly constrained scenarios that are addressable then it doesn't really help us as much with real world scenarios but it will be very cool to see how a lot of these daily phenomena connect to the things that we're talking about so let's see what else we had in the pieces and and um also if if axel is listening or wants to join for a dot 2 next week or for a future.³ i think there's probably many things we could ask and what brought you to write this paper and then where do you go after this one but pretty much the paper delivers the mic drop the coup de grace with the free energy minimizing bacteria which has bad priors or let's just say maladaptive priors and it minimizes free energy and makes the wrong decisions so that's where that section concludes then there's a jump back to the more philosophical framing so although free energy minimization is not sufficient for life there might be an entailment relation that goes the other way around if you are alive it might very well be because you did something like minimizing free energy that entailment relation is that which corresponds to the weak version of the free energy entailment relation so remember way back when when we were looking at the relationship between life and minimizing free energy this example has decimated the strong claim blown out make a youtube reaction video but the weak claim is actually supported because it seems like making adaptive decisions is consistent with free energy minimization just like it would be with making adaptive bayesian decisions but we can say that it's not sufficient to do so and then the name of the section free energy on a wing and a prior yuck yuck yuck anyone can give a comment on that but we just sort of were reminded of two winged creatures that we have talked about only a few weeks ago in 32 there's the winged snowflake and then the lepidoptera to a flame the moth to a flame over here and about how in this snowflake example so go to 32 to check it out in more detail but it's like the snowflakes with wings that act adaptively are going to persist and those are the ones that are going to be looking as if they've minimized free energy in selecting the correct action policies whereas the ones that don't exist just aren't if it makes bad decisions it's going to fail to exist it's an interesting section and it uses sort of evolutionary logic in a little bit of a different way because the evolutionary answer for everything is just sort of like that's what

was favored by evolution selection why does this organ have this shape it's like well that's what was selected on and so that's just that's just why it is that way even though we've talked about tim bergen and other things we know there's multiple y's but here we can now see the adaptivity of priors instead of relative fitness merely blue as we know more information here soon but the the adaptivity of the priors will be looking as if it has been minimizing free energy at the population level so whether we frame natural selection as increasing individuals with fitness or increasing the mean fitness of the population but not necessarily the relative fitness we can also do the same but rather with a malthusian exponent some other kind of economic fitness parameter we can think about it in terms of the free energy minimization yeah i keep falling off here so i'm gonna i'm gonna have to get past all the hidden stuff that you guys talked about when i wasn't uh staying on on the broadcast but what this points to me is interesting is back to the previous slide you talked about my fur rip dean please f in chat for dean he'll rejoin in just a few seconds sorry guys continue yeah is it a wing my first question is is it a wing as wing and then second what but does it still serve a function like could i use it to fly and so that's where the adaptive piece actually comes in for me so again there's two questions necessity and sufficiency right and and you can even see it in the representations is there enough information in the representation that i identify as wing and then the second part of it i can see it can carry a moth or i can see it can carry a snowflake that's the function that it serves and so that to me again is the minimum of two and making make sure that we don't blend those two things up because it's the fact that we hold both of them as discrete and each serving a different purpose that helps i'm gonna i'm gonna jump off prematurely because i tired of having to catch up and slow you guys down but i'll watch the video and seeing the point too thank you peace all right cool so we walk through much of the paper it's not a very long paper and it has just one core numerical example again with a bayesian walk through and then two variational bays one variational base that basically recapitulates the first bayesian example of an adaptive prior and then a second counter example that provides a maladaptive prior that is connected to evolution and to the idea that we're only going to see adaptive priors moving forward because if that model of recognition or that action model is maladaptive it will be seen less and less as time goes on blue so something that axel didn't mention in the paper or maybe um didn't mention or if i if he did mention it i didn't catch it but so the adaptive priors always fit a certain niche right and so like that you're only adaptive you know if you're like a thermophilic bacteria right like you're functional in a hot spring maybe not in like an ice patch right so i think that there's you know if you if you take the thermophilic

bacteria out of the hot environment and put it somewhere cold then it's it's maladaptive its priors are maladaptive but only because it's just not fit to the the correct niche environment and so i think that there's this you know the importance of of the niche environment and also it makes me wonder a lot in terms of evolutionary like conceptually like you know what if it's always we always think about like the best fit to the niche but but what if it's like the best generalist like is there like the evolutionary i mean i guess that's like the cockroach right like like the evolutionary generalists that can go anywhere be anywhere survive in any niche survive on anything like is that the best like how do how does that like like a prior that fits no niche or fits all niches how how does that like play into this evolutionary paradigm i mean i guess the cockroach will prevail good question maybe this is related to our discussion about context uh having the prior of 80 20 and then we observed alpha but we could have flipped that if it was all of a sudden dipped into the beta ocean then that prior would be as wrong as the maladaptive prior was in this paper whether or not it is adaptive as you point out it is absolutely about the statistical regularities of the niche and that's the fitness of the model fitness of the organism to generative process is the generative going to be fit to the generative process and then generative processes change things change out there in the yeah if the uh the work on my medics going back to dawkins and the the english lady that did a lot more development of that notion uh if that can be adapted to free energy that will give a lot of very detailed discussion of generation after generation of the co-evolution of adaptability and that which is adapting the the notion that the brain has been adapted to imitate and to remember and to pass on and to recast concepts into many different modalities so do what blue was asking agreed with that we can then just maybe look at the last few pieces in the future directions and then i think see if we have any ideas for things to discuss and point to and then other than that just invite everybody who's interested in listening to join us for point two because we've covered much in point one in zero so it's our last journal discussion of 2021 come join share how you got and then what action selections you're going to take in 22. the last section is free energy minimization as a historical scientific principle so that's a very fascinating idea it's about anticipatory systems yet it's a historical principle that's where we get into the difference between prediction and post-diction postdictive scientific statements are concerned with what must have been the case instead of what will be the case a statement such as the minimization of free energy may be necessary if not sufficient characteristic of evolutionary systems which was from the first and staff in 2007 is probably such a postdictive statement and then axel claims that that statement should be interpreted according to the weak

claim rather than according to the strong claim so interesting that even back then the necessity insufficiency was being used and arguably used properly so then what happened how did we get off the rails or how did the strong claim come to be makes a powerful argument for the fep as a postdictive model but that doesn't mean that it can't be used in an anticipatory way for anticipatory systems moving forward doing inference on the consequences of our action which are also in the future so how the uh past and the present and the future are linked that's more of a dot too how are the historical sciences anticipatory sciences and uh i don't know is there now sciences no that's what i call science 21 what else would people like to discuss in dot one i would like to maybe get into more about the living versus non-living and maybe talk about um you know this this idea of maintaining structural integrity like is that a quality of life or do things always maintain structural integrity and like does the fep apply when structural integrity is maintained and there's not life i mean it's like i mean we talk about like free energy minimization and chemistry all the time like you form the product that minimizes the free energy and so i know that there's some differences or some a lot of extrapolation that has to be done to get from thermodynamics to the fep or not extrapolation but like it's retrofit is maybe the right word um so i don't know i would like to maybe see some of where does the fep apply in situations that are not animated is it computationally is it relevant like when we're talking about robotics yeah that type of thing yeah and is animate the same as sentient like animate sentient inanimate not moving not animated in sentient can we just have an active model that doesn't have any affordances or where it only has one affordance so it's still doing a recognition model but then it's policy selection is extremely constrained or like so i mean i i don't even like to like use life and non-life because there are some things that like blur that boundary also like it gets into like well what is a virus is a virus alive like uh and a virus is very exploitative like it has dna like there's some things some qualities of viruses that make me feel like they're alive but they're technically not like what about if we just say dna containing but then it's like it's this test tube full of dna alive like i don't know it's hard and so instead of even sentient inanimate like alive and not alive i don't know yeah i i think the the paper's succinctness and clarity of argument hides a big question which is okay so if it's alive who told you that if we can't go from minimizing free energy to inferring what's alive then how did we get to here right or what is even alive mean like i don't think it's fully defined like if we can't fully define death i mean how do we fully define life right so i don't know it's this there's this big sticky i don't know weird area i mean we kind of instinctually know what is life and what is what is alive and what is not alive anyway but the i mean are the crystals in

bone alive well they're part of a bigger system that people usually don't have a problem saying is alive but then why should what people think matter especially if we believe in evolution and we think that our cognition is oriented not necessarily towards like ontological realism then why should it matter what one percent or 99 or even 100 of people agree upon or even if we zoom like way out and get super meta like how does life maintain the structural integrity of life that's an interesting one to think about like like because i mean it's dependent upon the niche environment which includes many non-alive things like the crystals that are inside of bone um or inside the inner ear which help us maintain balance right so there's all of those things yeah i think um following with this quote from the point of view of the fep the life cycle is that which corresponds to the thing whose integrity is maintained over evolutionary time so then the ants nest architecture is as much recapitulated as is the casing of the larva as is the larva itself how will we define life i guess we can just add that as a good you know just sort of just something quick to hit at the end of 32 or of 34.2 just you know what is life and then um hash rodingers question been answered i mean it's been a few years since 2018 and ram said it all but has it been answered what is life um okay so i think we will uh conclude there i hope if you're listening that you would like to join us in the dot 2 where we'll talk a little more about this paper and how we got here and then where we're going to go 22. anything else dave or blue okay thanks for the fun convo talk to you all later